

Commentary

Virtual Cell Challenge 2026: Benchmarking zero-shot generalization across cellular contexts

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The Virtual Cell Challenge returns in 2026 with a more demanding test of biological generalization: zero-shot prediction across multiple independent cellular contexts. Participants will build models to predict gene knockdown responses in a new Arc-generated dataset comprising unseen cell lines. The goal is to determine whether the best models can meaningfully close the gap between preclinical experimental predictions and human biology.

INTRODUCTION

Modern biomedical research often relies on preclinical systems, including animal models and cells in a dish, with the implicit assumption that the response measured in those assays will approximate the response in human patients. In reality, this assumption frequently fails. This translational gap is one reason that most candidate drugs entering clinical testing do not ultimately succeed.^{1,2} We can now perturb cells in culture and generate increasingly large perturbational datasets,^{3,4} but many of the cellular contexts that matter most for human biology remain difficult or impossible to perturb directly: rare cell types, primary cells that do not expand outside the body, diseased cells that cannot be cultured faithfully, and tissue states whose behavior depends on native microenvironments and interactions among multiple cell types. Therefore, even when such cells can be collected, removing them from the body changes the context we are trying to understand. Ultimately, the only setting in which we truly perturb human cells in their natural context is a clinical trial.

The Virtual Cell Challenge is designed to make this translational problem measurable.⁵ A virtual cell should not be judged only by whether it reproduces an experiment that could already be run cheaply and directly. It becomes valuable when it can use the experiments we can perform to make better predictions in the biological contexts where new experiments are

sparse, expensive, or unavailable. Success is not an exact reconstruction of the human response. Success is when a virtual cell prediction is meaningfully closer to the human response than the state-of-the-art preclinical experimental models alone.

FROM PERTURBATION PROFILING TO VIRTUAL CELLS

This goal also has an immediate experimental counterpart. For more than two decades, gene perturbation followed by gene expression profiling has been a key approach to understanding gene functions in cell biology. Public repositories such as the Gene Expression Omnibus^{6,7} are filled with knockdown, knockout, and other perturbation experiments followed by RNA sequencing (RNA-seq) or related expression readouts. A successful virtual cell model should be able to move this class of experiment *in silico*, both reducing the need to repeat similar perturbation-profiling assays across contexts and enabling researchers to explore perturbations, cell states, and combinations at a scale that is not experimentally practical (Figure 1A).

LESSONS FROM THE 2025 CHALLENGE

Last year, the inaugural Virtual Cell Challenge was launched as an open benchmark for perturbation prediction.⁵ Participants were asked to predict the transcriptional consequences of single-gene perturbations in H1 human embry-

onic stem cells using a purpose-built dataset and a shared evaluation framework. The response to the Challenge showed both the field's momentum and its current limits. More than 5,000 people registered across 114 countries, more than 1,200 teams submitted results, and more than 300 teams made final submissions. The strongest models improved on important capabilities such as perturbation discrimination and prediction of differentially expressed genes, but the results also made clear that current models are not yet consistently replacing experimental measurements across all evaluation criteria.^{8,9}

That outcome is precisely why the Challenge exists. The first year established a shared task, a high-quality benchmark dataset, and a community around rigorous evaluation. It also exposed where the field needs to mature: metric design, biological generalization, integration of statistical and deep learning approaches, and the ability to evaluate models in ways that reflect practical biological use rather than optimization of a single leaderboard score. The next step is to ask a more stringent question: can models predict perturbation responses in cellular contexts for which no challenge-specific perturbation training examples are provided (i.e., zero-shot prediction)?

THE 2026 VIRTUAL CELL CHALLENGE

The 2026 Virtual Cell Challenge builds on these lessons by focusing on multi-

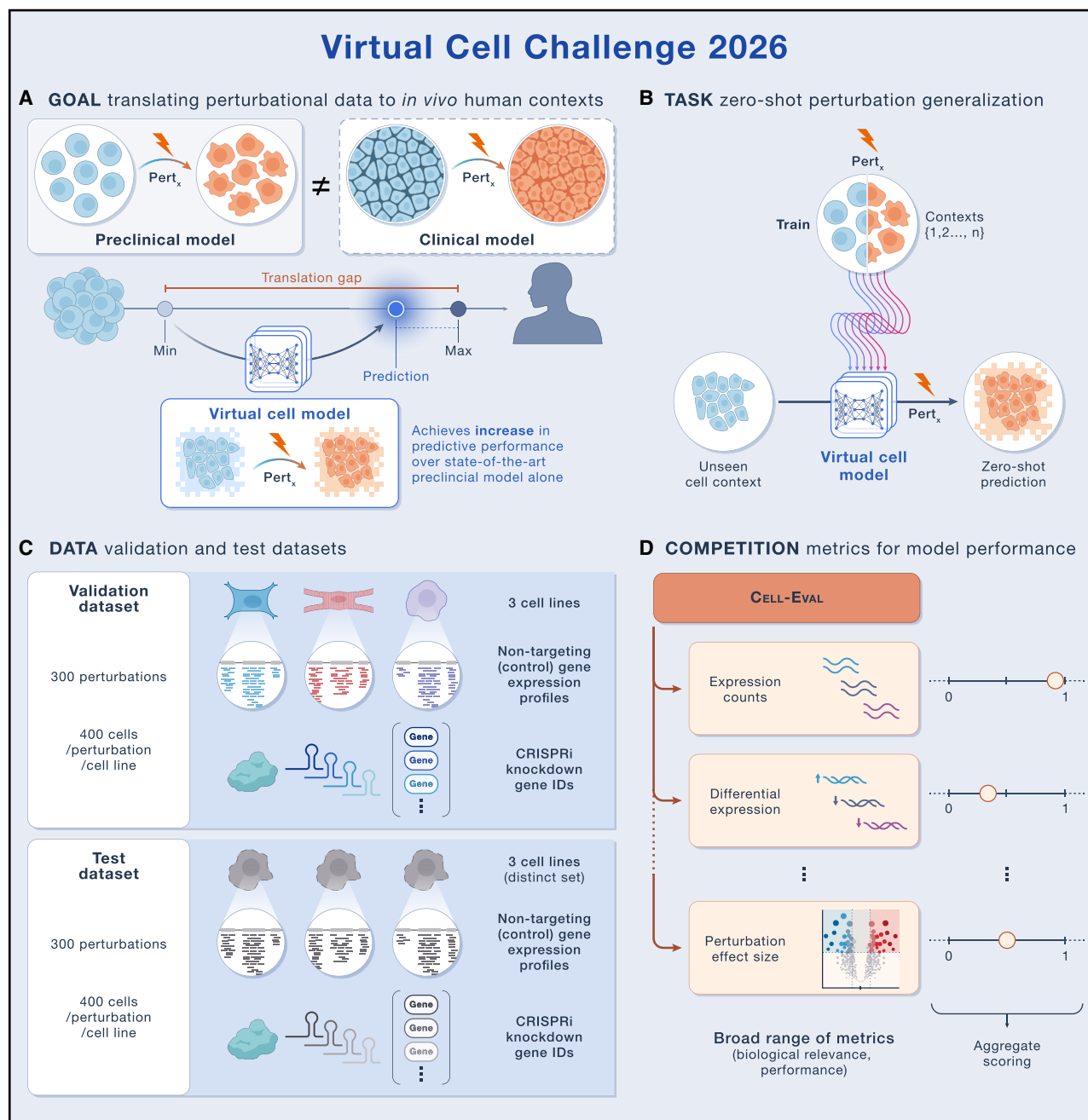


Figure 1. The 2026 Virtual Cell Challenge

(A) A key goal of virtual cells for this Challenge is that these predictive models are more accurate than preclinical experimental models.

(B) The task for the 2026 Virtual Cell Challenge is to build a model capable of multi-context generalization and zero-shot prediction. Models will need to predict the post-perturbation expression profiles that would be measured after CRISPRi knockdown followed by 10x Flex single-cell profiling based on profiles from cells expressing non-targeting guides together with the gene identifiers for CRISPRi knockdowns to be predicted.

(C) The 2026 Virtual Cell Challenge is built on six distinct CRISPRi cancer cell line datasets. Three of these will be used during the validation phase, while the remaining three will be reserved for the testing phase.

(D) Model performance will be evaluated using a suite of metrics.

context generalization and zero-shot prediction. Unlike the 2025 Challenge, there will not be a challenge-specific training set. Participants are encouraged to use

any appropriate data and modeling strategy, but the Challenge itself will test whether those models can generalize to new cellular contexts without matched

perturbation-response examples from those contexts (Figure 1B). At Arc Institute, we have performed Perturb-seq experiments on six cell lines originating

from different tissues using CRISPR interference (CRISPRi) and 10x Flex single-cell RNA-seq (scRNA-seq).¹⁰

At the start of the Challenge, participants will receive validation inputs for three of these cell lines. For each cell line, we will release profiles from cells expressing non-targeting guides together with the gene identifiers for CRISPRi knockdowns to be predicted (Figure 1C). Models must use these inputs to predict the post-perturbation expression profiles that would be measured after CRISPRi knockdown followed by 10x Flex single-cell profiling.

This design intentionally separates access to the target context from access to perturbation outcomes in that context. The non-targeting guide profiles define the basal state of each cell line, while the perturbation identities define the interventions to be simulated. The withheld measurements define the ground truth. In this setting, a successful model must infer how a perturbation response transfers across contexts without being given matched perturbation-response training data from the Challenge itself.

EVALUATION

The live leaderboard will be based on the validation set and ranked using a subset of metrics from Cell-Eval,¹¹ as in the inaugural Challenge. The final test set will follow a similar structure, but it will use three different cell lines from the validation set. During the final week, participants will receive the test inputs and submit final predictions without receiving feedback on performance. Final rankings will be based on this held-out evaluation.

The 2026 Challenge will continue to use rigorous, transparent evaluation. Cell-Eval provides a framework for measuring complementary aspects of perturbation prediction, including whether models recover differentially expressed genes, distinguish perturbation effects, and approximate measured expression profiles. The first Challenge showed that no metric fully captures model quality, consistent with other observations.¹² This year, final scoring will use a reference-normalized aggregate across six metrics (Figure 1D). This preserves a broad evaluation of model behavior while reducing the incentive to optimize for a single metric.

COMMUNITY ENGAGEMENT

We hope this year's Challenge will continue to attract a broad community. Machine learning researchers are needed to build models that generalize across high-dimensional biological data without relying on task-specific training examples. Computational biologists are needed to define what meaningful generalization looks like and where models fail biologically. Experimental and translational researchers are needed because the central problem is not abstract prediction but rather the ability to replace, prioritize, or redirect real perturbation-profiling experiments. The Challenge is designed to make these perspectives interdependent. Universities, companies, research institutions, and individual teams will all make important contributions in this ecosystem.

CONCLUSION

The Virtual Cell Challenge is an annual, open effort to define the standards by which virtual cell models should be tested. Each year, the datasets and evaluation framework will evolve with the field. In 2026, the bar is higher, from adaptation within a measured context to zero-shot prediction across multiple contexts. If the field can learn to move a foundational perturbation assay *in silico*, it will create the conditions for an AlphaFold moment in cellular modeling—not because experiments become unnecessary, but because models can help researchers decide which experiments matter most.

This framing reflects our working definition of a virtual cell as a predictive model of cellular state and response: a model that takes a starting cellular context and a perturbation, then simulates the molecular consequences of that perturbation well enough to guide biological reasoning and experimental prioritization. The immediate readout may be gene expression, differentially expressed genes, or another measurable molecular phenotype, but the broader goal is to build models that can support discovery: proposing hypotheses, prioritizing interventions, and identifying where experimental validation is most likely to be informative.

DECLARATION OF INTERESTS

D.P.B. is a Google Advisor. H.G. is a cofounder of Network Bio, Tahoe Therapeutics, and Therna Therapeutics; serves on the board of directors at Network Bio; and is a board director of Verge Labs. P.D.H. is a cofounder of Monet AI, Terrain Biosciences, and Stylus Medicine; serves on the board of directors at Stylus Medicine; is a board observer at EvolutionaryScale and Terrain Biosciences; is a scientific advisory board member at Arbor Biosciences and VedaBio; and is an advisor to NFDG, Varda Space, and Vial Health. L.A.G. is a cofounder of nChroma Bio and a scientific advisory board member of Myllia Biotechnology.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cell.2026.08.004>.

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Supplemental information

Virtual Cell Challenge 2026:

**Benchmarking zero-shot generalization
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